

Goal: Find Best Circuit Architecture

(a) SuperCircuit Method (e.g., QuantumNAS)

Device-Dependent & Training-Heavy

Candidate SubCircuit

SuperCircuit Training

Noisy Evaluation
on Specific Device D

Fitness: Noisy Loss
(Requires Dataset)

Best Circuit for Device D

Cost Legend

Green → Low / Negligible Cost

Yellow → Medium Cost

Red → High Cost

(b) Proposed Method (Ours)

Device-Agnostic & Training-Free

Candidate SubCircuit

No Training Required
(Random Parameters)

Noise-Free Expressibility
(KL to Haar)

Fitness: Structural Proxy
(No Dataset Needed)

Best Architecture
(Deployable to Any Device)